MERN STACK

TRAINING TR-103 REPORT DAY 9

4 July 2026

Overview:

Today I started learning **JavaScript.** I understood the fundamentals of **JavaScript**, the programming language used to make web pages interactive and dynamic. I learned different ways to add JavaScript to HTML, explored variables, operators, conditional statements, and the parseInt() function. I also practiced using console.log() for debugging and writing simple JavaScript programs.

Learning Objectives:

- Understand the basics of JavaScript.
- Learn different ways to add JavaScript to HTML.
- Understand JavaScript variables (var, let, and const).
- Learn different types of JavaScript operators.
- Understand conditional statements in JavaScript.
- Learn the use of console.log() and parseInt().

Introduction to JavaScript

JavaScript is a programming language used to make web pages interactive and dynamic. It supports both client-side and server-side development and allows developers to create interactive web applications.

Features:

- Interpreted language.
- Dynamically typed.
- Single-threaded with asynchronous support.

Adding JavaScript to HTML :

JavaScript is added to an HTML document using the <script> tag.

Ways to add JavaScript:

- **Inline JavaScript** – Written inside HTML elements using event attributes.
- **Internal JavaScript** – Written inside the <script> tag in an HTML file.
- **External JavaScript** – Written in a separate .js file and linked to HTML.

Advantages of External JavaScript:

- Improves code organization.
- Allows code reuse.
- Easy to maintain.
- Improves page loading through browser caching.

console.log()

The console.log() function is used to display messages or output in the browser console. It is mainly used for testing and debugging JavaScript programs.

Example:

```
console.log("Hello, World!");
```

JavaScript Variables

Variables are used to store data values in JavaScript. Since JavaScript is dynamically typed, there is no need to specify the data type.

Types of Variables:

- **var** – Function-scoped and can be redeclared.
- **let** – Block-scoped and can be updated.
- **const** – Block-scoped and cannot be reassigned.

```
let a = 10;
let b = 10;
b=30;
const c =40;
var d = "Jeeya Thapar"
var e = 88;
```

JavaScript Operators

Operators are symbols used to perform different operations on variables and values.

Types of Operators:

- **Arithmetic Operators** – +, -, *, /, %
- **Assignment Operators** – =, +=, -=, *=, /=
- **Comparison Operators** – ==, ===, !=, >, <, >=, <=
- **Logical Operators** – &&, ||, !
- **Unary Operators** – ++, --, typeof
- **String Operators** – +, +=

TASK -1

(To perform operators)

```
// OPERATORS

//Airthmetic operator
        console.log("-----Arithmetic Operators----")
        let a = 5
        let b = 5
        console.log("Sum of two number is:", a+b)
        console.log("Difference of two number is:",a-b)
        console.log("Product of two number is:",a*b)
        console.log("Division of two number is:",a/b)

//Uniary operator
        console.log("-----Uniary Operator----")
        console.log("Increment of a number: ",a++)
        console.log("Decrement of a number: ",++a)

// Assignment operators
        console.log("-----Assignment Operator----")
        let num = 5
        num +=5
        console.log(num)
        num-=2
        console.log(num)

//Comparison Operators -------result must be TRUE OR FALSE
        console.log("-----Comparison Operator----")
        console.log(a==b)
        console.log(a<b)
        let c= 5
        let d = 5
        console.log(c===d)

//Logical Operator
        console.log("-----Logical Operator----")
        let aa = 10;
        let bb = 20;

        console.log(aa > bb || aa < bb);   // true
        console.log(aa > bb || aa < 5);   // false
```

```
-----Arithmetic Operators----
Sum of two number is: 10
Difference of two number is: 0
Product of two number is: 25
Division of two number is: 1
-----Uniary Operator----
Increment of a number:  5
Decrement of a number:  7
-----Assignment Operator----
10
8
-----Comparison Operator----
false
false
true
-----Logical Operator----
true
false
```

Conditional Statements

Conditional statements are used to make decisions based on conditions.

Types of Conditional Statements:

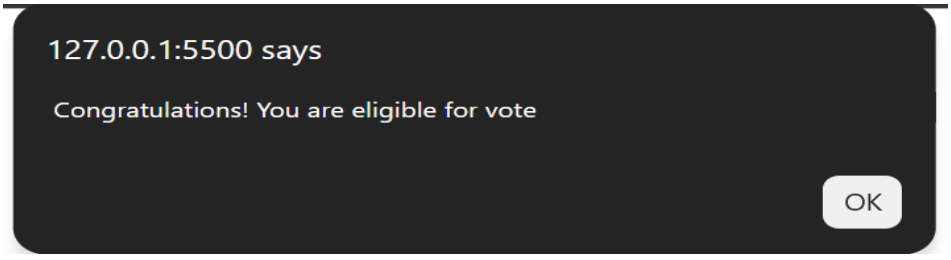
- **if** – Executes code if the condition is true.
- **if...else** – Executes one block if true, otherwise another block.
- **if...else if...else** – Checks multiple conditions.
- **switch** – Executes code based on different cases.

TASK -2

(To check a person is eligible for vote or not)

```
let age =(prompt("Enter your age:"))
  if(age>=18){
      alert(" Congratulations! You are eligible for vote")
  }
  else{
      alert(" Oops! You are not eligible for vote")
  }
```

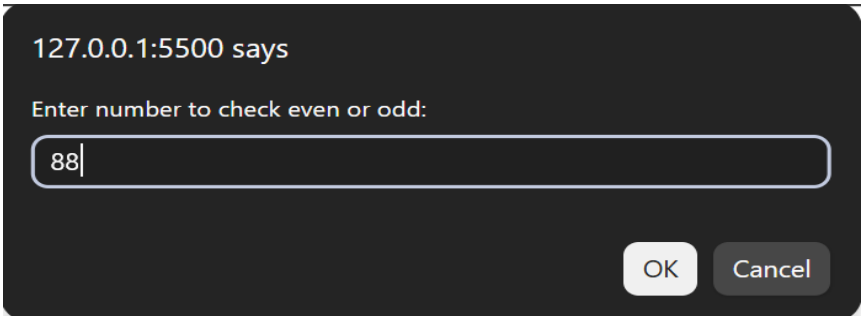


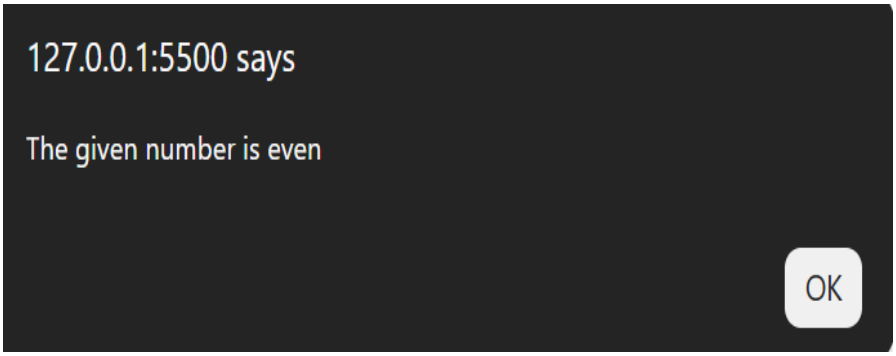


TASK –3

(To check a given number is even or odd)

```
let num = (prompt("Enter number to check even or odd:"))
if(num /=2){
    alert("The given number is even")
}
else{
    alert("The given is Odd")
}
```





parseInt() Function

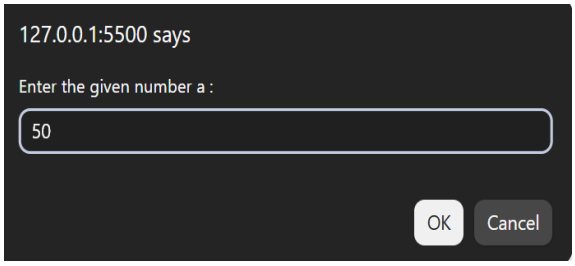
The parseInt() function converts a string into an integer (whole number). If the value is not a valid number, it returns NaN (Not a Number).

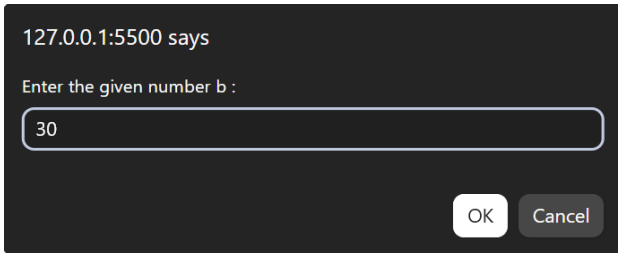
Features:

- Converts a string to an integer.
- Ignores decimal values.
- Returns **NaN** for invalid input.

TASK –4

```
let a = parseInt(prompt("Enter the given number a :"))
let b = parseInt(prompt("Enter the given number b :"))
console.log("Sum of two number is:", a+b)
console.log("Difference of two number is:",a-b)
console.log("Product of two number is:",a*b)
console.log("Division of two number is:",a/b)
```






```
Sum of two number is: 80
Difference of two number is: 20
Product of two number is: 1500
Division of two number is: 1.6666666666666667
```

Conclusion:

By the end of Day 9, I gained a basic understanding of JavaScript and learned how to add it to HTML pages. I also explored variables, operators, conditional statements, `console.log()`, and `parseInt()`, which helped me understand the fundamentals of writing interactive JavaScript programs.